322304

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Figures to the right indicate full marks.

(5) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Answer any FIVE of the following : 10

(a) Draw a diagram of container used for collection of blood.

(b) Define quality control.

(c) Draw a diagram of container used for collection of urine.

(d) Write different pH ranges.

(e) Write any two uses of Laminar air flow.

(f) Define spectroscopy.

(g) Define Chromatography.

2. Attempt any THREE of the following : 12

(a) Give statements of Beer's and Lambert's law.

(b) Explain determination of optical density of given solution by calorimeter.

(c) Describe principle and components of electrophoresis.

(d) Explain principle and procedure to determine blood glucose by using glucometer.

3. Answer any THREE of the following : 12

(a) Draw a neat well labelled diagram of electrophoresis apparatus.

(b) Explain working and uses of electrophoresis technique in medical diagnosis.

(c) Describe components, care and uses of spectrophotometer.

(d) Draw a neat well labelled diagram of thin layer chromatography.

4. Attempt any THREE of the following : 12

(a) Describe step by step procedure for collection of blood.

(b) Draw and explain laboratory safety chart.

(c) Describe components along with their functions in flame photometer.

(d) Describe the working principle of and wavelength of UV spectrophotometer.

(e) Explain uses of flame photometer.

5. Attempt any TWO of the following : 12

(a) Explain principle, components, care and uses of Incubator.

(b) Enlist types of quality control. Explain leavy Jennings and cusum chart.

(c) Explain principle, component, care and uses of centrifuge.

6. Attempt any TWO of the following : 12

(a) Explain principle, care and components of analyzer.

(b) Explain the principle, component, care and uses of Autoclave.

(c) Explain principle, components, working, care and uses of thin layer chromatography.